

WEB DEVELOPMENT

A structural reference sheet defining the discipline, its layers, its build process, and the vocabulary you'll need on site.

◆Definition

WHAT IT IS

Web development is the practice of building and maintaining **websites and web applications** — the process of turning a design or an idea into something that runs inside a browser, is reachable over the internet, and responds to what a user does.

FULL DEFINITION

Web development is the discipline of designing, building, testing, and maintaining software that is delivered and consumed through a web browser. It spans everything a user sees and clicks (the **frontend**), the logic and data that power it behind the scenes (the **backend**), and the infrastructure that keeps it running, secure, and available at scale (**DevOps**). Unlike a desktop app, a web application is built to be reached instantly by anyone with a link — no install required — which is precisely what makes it both powerful and demanding to build correctly.

◆Structure

THE CORE LAYERS

Every web build rests on the same four load-bearing layers, no matter the framework you reach for.

L-01

FRONTEND

What the user sees and touches. Built with **HTML** (structure), **CSS** (style), and **JavaScript** (behavior). Frameworks like React, Vue, or Svelte sit on top of this.

L-02

BACKEND

The server-side logic — handles requests, business rules, authentication. Written in languages like **Node.js**, **Python**, **Go**, **Java**, exposed through APIs.

L-03

DATABASE

Where data persists. **SQL** (Postgres, MySQL) for structured relational data, **NoSQL** (MongoDB, Redis) for flexible or fast-access data.

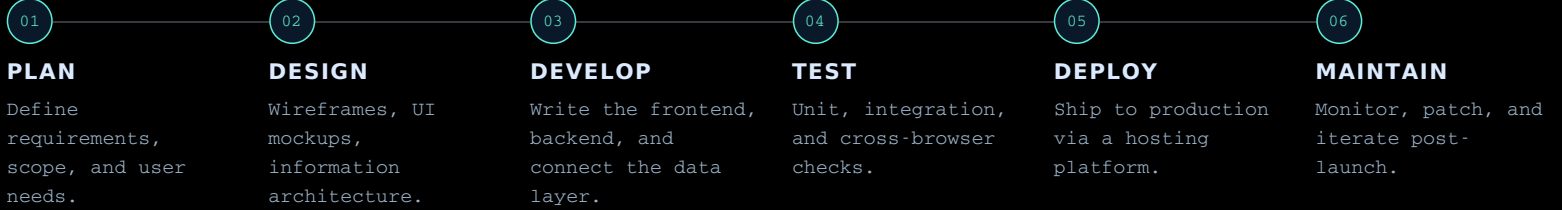
L-04

INFRASTRUCTURE

Hosting, deployment, and monitoring. Covers **servers**, **CDNs**, **CI/CD pipelines**, and cloud platforms like AWS, Vercel, or GCP.

BUILD LIFECYCLE

The sequence a real project moves through, start to maintenance.



FIELD GLOSSARY

HTML

HyperText Markup Language — the skeleton that structures content on a page.

CSS

Cascading Style Sheets — controls layout, color, spacing, and visual design.

JAVASCRIPT

The language that makes pages interactive — handles logic, events, and dynamic updates.

API

Application Programming Interface — the contract that lets frontend and backend talk to each other.

RESPONSIVE DESIGN

Layouts that adapt cleanly across phone, tablet, and desktop screens.

FRAMEWORK

A pre-built toolkit (React, Django, Express) that speeds up common patterns.

VERSION CONTROL

Tracking code changes over time — almost always done with Git.

DEPLOYMENT

The act of pushing a working build live to a server users can reach.